

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 11

BIOTECHNOLOGY

BIOTECHNOLOGY

DEFINITION

- Biological Technology is the extension of technology into biology. It is the use of microbial, animal and plant cells or enzymes to make, break or transform materials.
- Biotechnology deals with industrial scale production of biopharmaceuticals and biologicals using genetically modified microbes, plants and animals.



Biotechnology is the integration of organic chemistry, biochemistry, microbiology, chemical engineering etc.

APPLICATIONS OF BIOTECHNOLOGY

INSULIN PRODUCTION

- Insulin is a hormone made by Pancreas that allows the body to use sugar (glucose) from the food we eat for energy or to store for future use.
- Insulin helps to keep the blood sugar level from getting too high which is called Hyperglycemia or too low which is Hypoglycemia.



APPLICATIONS OF BIOTECHNOLOGY

FUNCTION OF INSULIN

- When insulin is released into the blood stream, it attaches to the cells and signals cells to absorb sugar from the blood.
- Insulin is like a key which unlocks the cell to allow sugar to enter into the cell.
- Insulin acts as a drug to treat diabetes.
- Previously production of insulin was done through animal process. It was very cruel and expensive as a large number of animals had to be killed to produce a few grams of insulin.
- Insulin from an animal source, also caused some patients to develop allergy and other types of reactions.



APPLICATIONS OF BIOTECHNOLOGY

FUNCTION OF INSULIN

- The chemical method of producing insulin is also difficult and expensive.
- Today biotechnology made it very simple. A synthetic gene responsible for producing insulin is prepared and linked to E. Coli or Yeast (through Recombinant DNA Technology) and then insulin is obtained.
- The incidence of diabetes is increasing at an alarming rate and it has been estimated that the number of diabetic patients worldwide would increase to 300 million by 2025.
- Recombinant DNA Technology (rDNA) addresses this challenge.



APPLICATIONS OF BIOTECHNOLOGY

HORMONAL PRODUCTION

- Hormones are chemical compounds secreted by endocrine glands.
- They regulate vital body functions.
- They are used to curing certain diseases caused by their deficiency.
- Artificial synthesis of hormones is very difficult and expensive. But techniques like Recombinant DNA Technology and Gene Cloning are the solution to this problem.
- Example – Two hormones **Somatostatin** and **Somatotrophin** are produced by Recombinant DNA Technology.

APPLICATIONS OF BIOTECHNOLOGY

HORMONAL PRODUCTION

SOMATOSTATIN

- Also called **Growth Hormone Inhibiting Hormone**.
- It is secreted from the Duodenum and Pancreas.
- It affects several areas of the body. It regulates the secretion of hormones coming from the Pituitary gland (growth hormone, thyroid stimulating hormone etc.) and inhibits the secretion of pancreatic hormones (glucagon and insulin).

SOMATOTROPHIN

- It is growth hormone important for the growth of bones and muscles.
- It is secreted by Pituitary gland.

APPLICATIONS OF BIOTECHNOLOGY

ENZYME TECHNOLOGY

- It refers to the isolation, purification and production of enzymes.
- Enzymes are complex organic molecules present in cells. They act as catalyst for all the biochemical reactions happening in the cell.
- Enzymes are present in only living cells.
- But today it is possible to separate, isolate and use enzymes in vitro. (In vitro – taking place in a test tube or anywhere outside the living organism.)
- This capability of enzymes to catalyze reactions outside a cell has led to a separate branch of science known as Enzyme Technology.

APPLICATIONS OF BIOTECHNOLOGY

ENZYME TECHNOLOGY

The advantages of this process are

- a) Unethical cruelty is avoided.
- b) Microorganisms are available in abundance compared to animals.
- c) The amount of enzymes produced is more.
- d) Pure and stable enzymes are extracted.

Artificial enzymes are also produced using Gene Cloning technique.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588